EWB COURSES

EWB CLOUD LABS
STUDENT PROJECT REPORT

BUILD / TEST / EXPLAIN

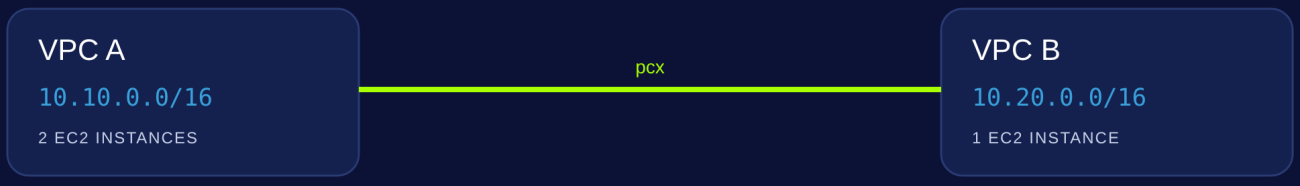
AWS

VPC Peering

Connect two VPCs.

Prove private connectivity.

AWS VPC Peering Between Two Amazon VPCs



2 VPCs / 3 EC2 INSTANCES / 1 PRIVATE CONNECTION

AWS ARCHITECT ASSOCIATE

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BATCH	**Batch 1**
PROJECT TITLE	**AWS VPC Peering Between Two Amazon VPCs**
AWS REGION	**Asia Pacific (Mumbai), ap-south-1**
COMPLETION DATE	**23 October 2026**

LEARN. ELEVATE. EMPOWER.

01 / OVERVIEW

Project objective

Connect two VPCs and check that private communication works.

In this project I connected two Amazon VPCs in the Asia Pacific (Mumbai) Region (ap-south-1) using a VPC peering connection, so that an EC2 instance in one VPC can talk to an instance in the other using the private IPv4 address assigned to it inside its VPC. I set up the route tables and security groups for both sides and tested the connection from the command line with ping and SSH.

What VPC peering is

VPC peering is a private, one-to-one network link between two VPCs. Traffic between them stays on the AWS network and does not need an internet gateway. Creating the peering connection is not enough on its own. Each VPC also needs a route to the other VPC's CIDR block that points to the peering connection, and the security groups must allow the traffic. Peering is not transitive, and the two VPCs cannot have overlapping CIDR blocks.

What I did

- Created two VPCs with CIDR blocks that do not overlap: 10.10.0.0/16 (ewb-vpc-a) and 10.20.0.0/16 (ewb-vpc-b).
- Created three subnets in ap-south-1a: ewb-a-public (10.10.1.0/24), ewb-a-private (10.10.2.0/24) and ewb-b-private (10.20.1.0/24).
- Created three custom route tables, one per subnet: ewb-rt-a-public, ewb-rt-a-private and ewb-rt-b-private.
- Created the peering connection ewb-a-b-peer (pcx-00d88e5dafce78b7e), which shows Active.
- Added a route in each route table that points to the other VPC's CIDR through the peering connection.
- Launched three t3.micro instances: ewb-bastion (public IP), ewb-app-a and ewb-app-b (no public IP), and set up security group rules for SSH and ICMP.
- Logged in by SSH to ewb-app-a through the bastion (ProxyJump) and pinged ewb-app-b by its private IP: 4 sent, 4 received.
- Cleaned up afterwards: terminated the instances, deleted the peering connection, and checked that the Elastic IPs and the project's subnets, route tables, security groups, internet gateway and VPCs are gone.

About my CIDR blocks
I used 10.10.0.0/16 and 10.20.0.0/16. Both are inside the RFC 1918 private range 10.0.0.0/8 and they do not overlap, which VPC peering requires.

About the evidence
Every value in this report comes from my screenshots. If a value is not visible in a screenshot, I have written that. Things the workbook asks for that I did not capture are marked 'Not captured'. Appendix B lists all of them.

02 / ARCHITECTURE

The network I built

Two VPCs, three EC2 instances and one peering connection.

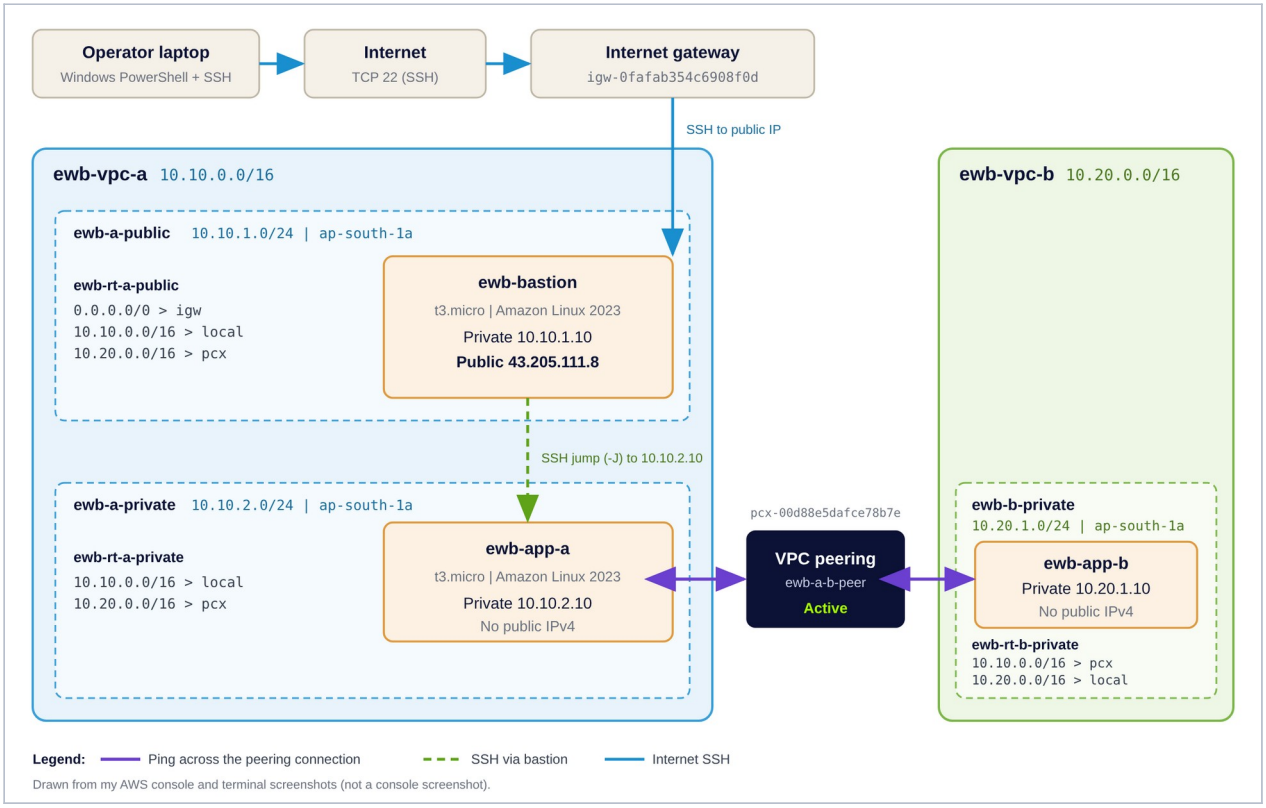


Figure 1. Architecture diagram drawn from my screenshots (not a console screenshot).

Workbook design compared with my build

The workbook expects a separate bastion host in the public subnet of VPC A, two application hosts with no public IP, and three custom route tables. My build follows that design. The table shows each item next to what I built.

Item	Workbook design	What I built
VPC CIDRs	10.10.0.0/16 and 10.20.0.0/16	10.10.0.0/16 (ewb-vpc-a) and 10.20.0.0/16 (ewb-vpc-b), no overlap
Region	Example: ap-south-1	ap-south-1 (Asia Pacific, Mumbai)
EC2 instances	3: bastion, App-A, App-B	3: ewb-bastion, ewb-app-a, ewb-app-b
Bastion	Separate bastion in the VPC A public subnet	ewb-bastion in ewb-a-public (private 10.10.1.10, public 43.205.111.8)
Private subnet in VPC A	ewb-a-private with its own route table	ewb-a-private (10.10.2.0/24) with ewb-rt-a-private
Route tables	3 custom tables with explicit associations	ewb-rt-a-public, ewb-rt-a-private, ewb-rt-b-private, each associated with one subnet
Instance type / OS	t3.micro suggested, Amazon Linux 2023	t3.micro, Amazon Linux 2023 (shown in the terminal banner)
SSH method	ProxyJump with one key pair	ssh -A with -J through the bastion; one key loaded in the SSH agent (Figure 20)

Item	Workbook design	What I built
ICMP rule on destination	Echo request from a single /32	All ICMP - IPv4 from a single /32: the other app instance's IP
Naming	Lowercase ewb-* names	ewb-* names (for example ewb-a-b-peer)

03 / RESOURCE INVENTORY

Resources and IDs

Names, IDs and IP addresses used in this project.

Resource	Name	ID / value
VPC A	ewb-vpc-a	vpc-0baaa64b599a9564a | 10.10.0.0/16
VPC B	ewb-vpc-b	vpc-0edfc5a72b7f3fa9e | 10.20.0.0/16
VPC A public subnet	ewb-a-public	subnet-01c584803255d3401 | 10.10.1.0/24 | ap-south-1a
VPC A private subnet	ewb-a-private	subnet-07d7cd4a11c071c32 | 10.10.2.0/24 | ap-south-1a
VPC B private subnet	ewb-b-private	subnet-0a29f322b78b51473 | 10.20.1.0/24 | ap-south-1a
Internet gateway (a)	Name not shown	igw-0fafab354c6908f0d | target of 0.0.0.0/0 in ewb-rt-a-public
VPC A public route table	ewb-rt-a-public	rtb-03bb7a349ca805781
VPC A private route table	ewb-rt-a-private	rtb-04b5d108613636f58
VPC B private route table	ewb-rt-b-private	rtb-0f2c292903f88f77e
VPC A main route table	Unnamed	rtb-033be1caa1aa2f5b9
VPC B main route table	Unnamed	rtb-0f9ab50c83d64617f
Peering connection	ewb-a-b-peer	pcx-00d88e5dafce78b7e | Active
Security group 1 (b)	Name not shown	sg-0ef812dc16c705a9d | SSH from one public /32
Security group 2 (b)	Name not shown	sg-0e267dfedbcba142f | rules for ewb-app-a
Security group 3 (b)	Name not shown	sg-02ab0878c5cbfeaf9 | rules for ewb-app-b
EC2 instance, bastion	ewb-bastion	i-07a957f5791d232bc | private 10.10.1.10 | public 43.205.111.8
EC2 instance, app A	ewb-app-a	i-09a5f9baa26eaafb8 | private 10.10.2.10 | no public IPv4
EC2 instance, app B	ewb-app-b	i-01695738463794277 | private 10.20.1.10 | no public IPv4

Notes on the inventory

(a) The internet gateway appears only as the target of the 0.0.0.0/0 route (Figure 9), so its name and attached VPC are not visible in my screenshots.

(b) The security group screenshots show the group ID but not the group name. I matched the groups to instances from their rule sources: the group with SSH from one public /32 is for the bastion, and the two groups with ICMP from the other app instance are for ewb-app-a and ewb-app-b.

The VPC list also shows a third VPC (172.31.0.0/16, vpc-084a7dc8cd5639b2a). That is the account's default VPC and is not part of this project. All three project subnets are in ap-south-1a (aps1-az1).

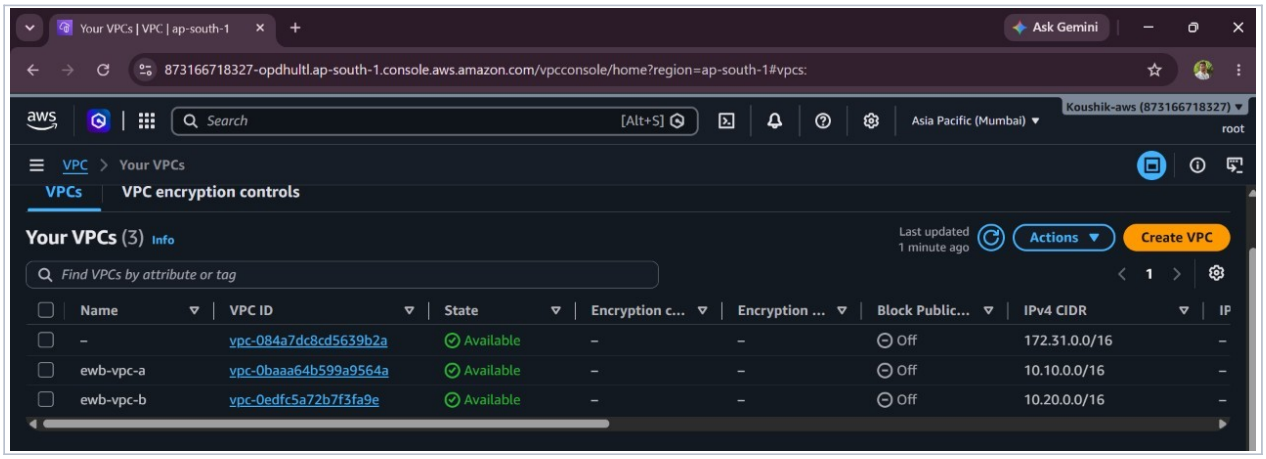


Figure 2. Your VPCs list (ap-south-1): ewb-vpc-a (10.10.0.0/16) and ewb-vpc-b (10.20.0.0/16), both Available. The default VPC (172.31.0.0/16) is also listed.

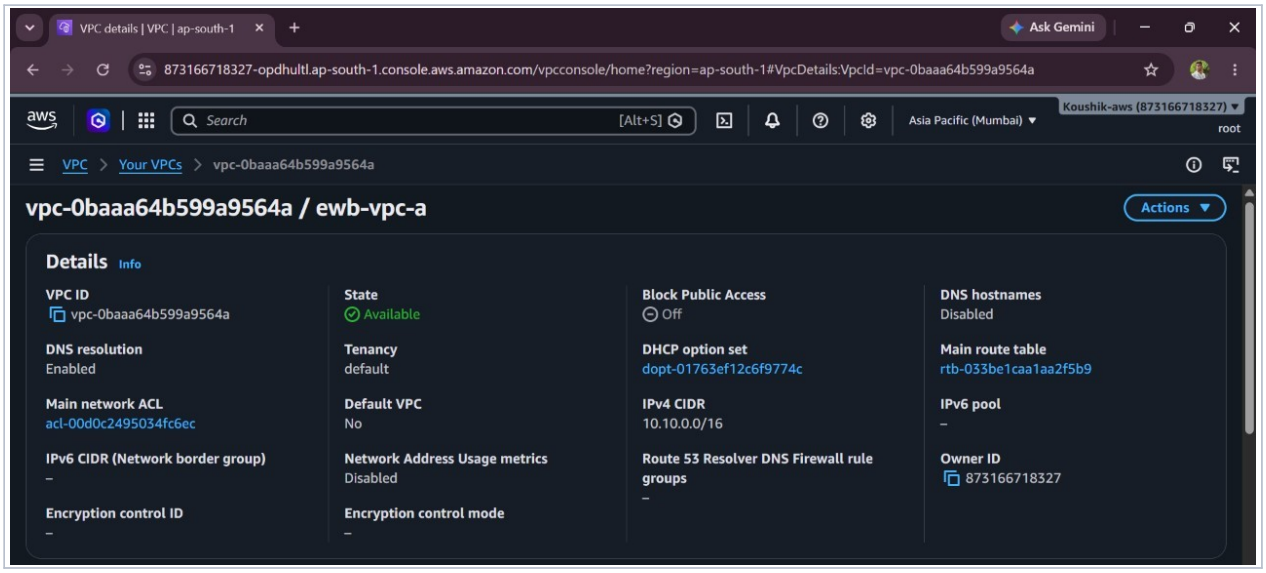


Figure 3. ewb-vpc-a (vpc-0baaa64b599a9564a): 10.10.0.0/16, Available, DNS resolution Enabled, DNS hostnames Disabled, main route table rtb-033be1caa1aa2f5b9.

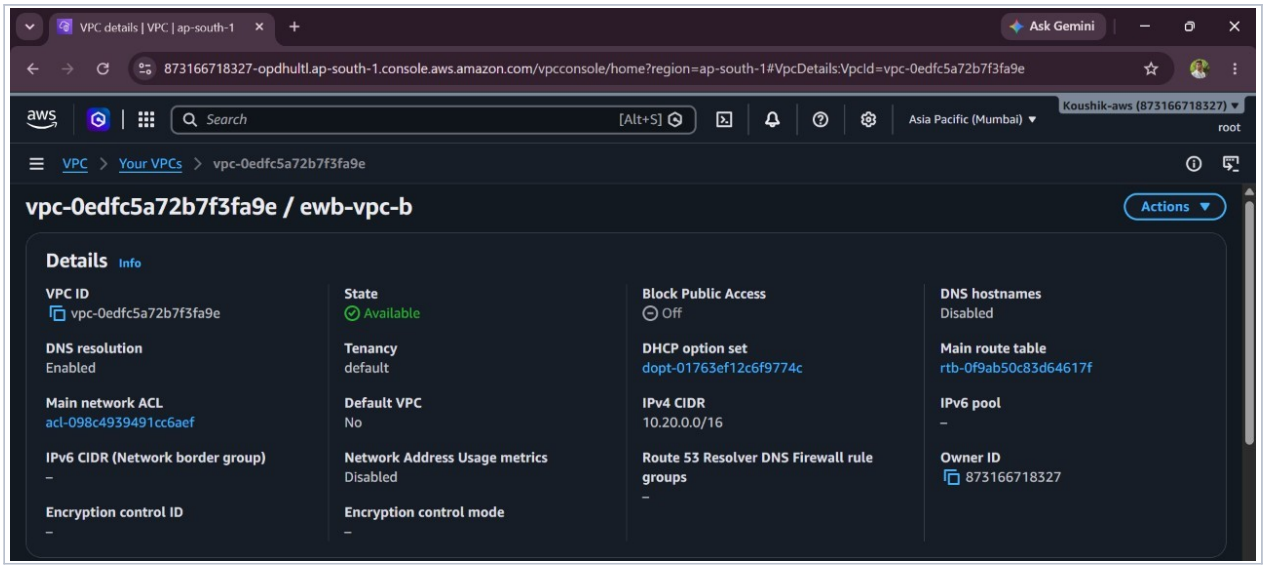


Figure 4. ewb-vpc-b (vpc-0edfc5a72b7f3fa9e): 10.20.0.0/16, Available, DNS resolution Enabled, DNS hostnames Disabled, main route table rtb-0f9ab50c83d64617f.

Subnets and route tables

The routes on both sides let traffic and replies use the peering connection.

Subnets

Subnet	ID	IPv4 CIDR	Auto-assign public IPv4	Route table
ewb-a-public (VPC A)	subnet-01c584803255d3401	10.10.1.0/24	No	ewb-rt-a-public (rtb-03bb7a349ca805781)
ewb-a-private (VPC A)	subnet-07d7cd4a11c071c32	10.10.2.0/24	No	ewb-rt-a-private (rtb-04b5d108613636f58)
ewb-b-private (VPC B)	subnet-0a29f322b78b51473	10.20.1.0/24	No	ewb-rt-b-private (rtb-0f2c292903f88f77e)

All three subnets are in ap-south-1a (aps1-az1), are Available, and each shows 250 available IPv4 addresses. The bastion is the only instance with a public IPv4 address, and it gets it from the instance launch setting, not from the subnet (auto-assign is No on the subnet).

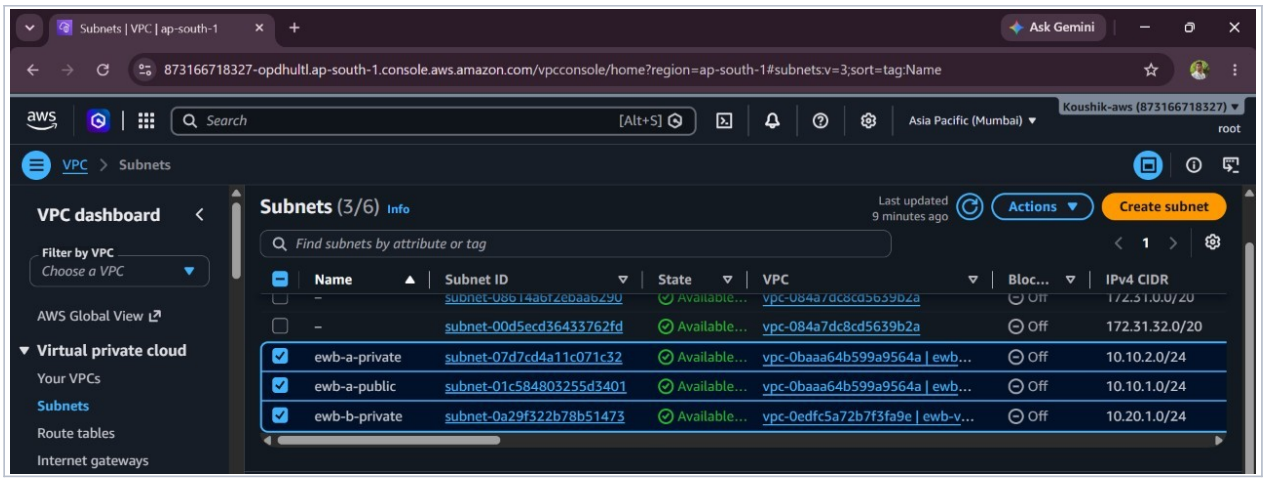


Figure 5. Subnets list with ewb-a-private, ewb-a-public and ewb-b-private selected (10.10.2.0/24, 10.10.1.0/24, 10.20.1.0/24). The two default-VPC subnets above them are not part of the project.

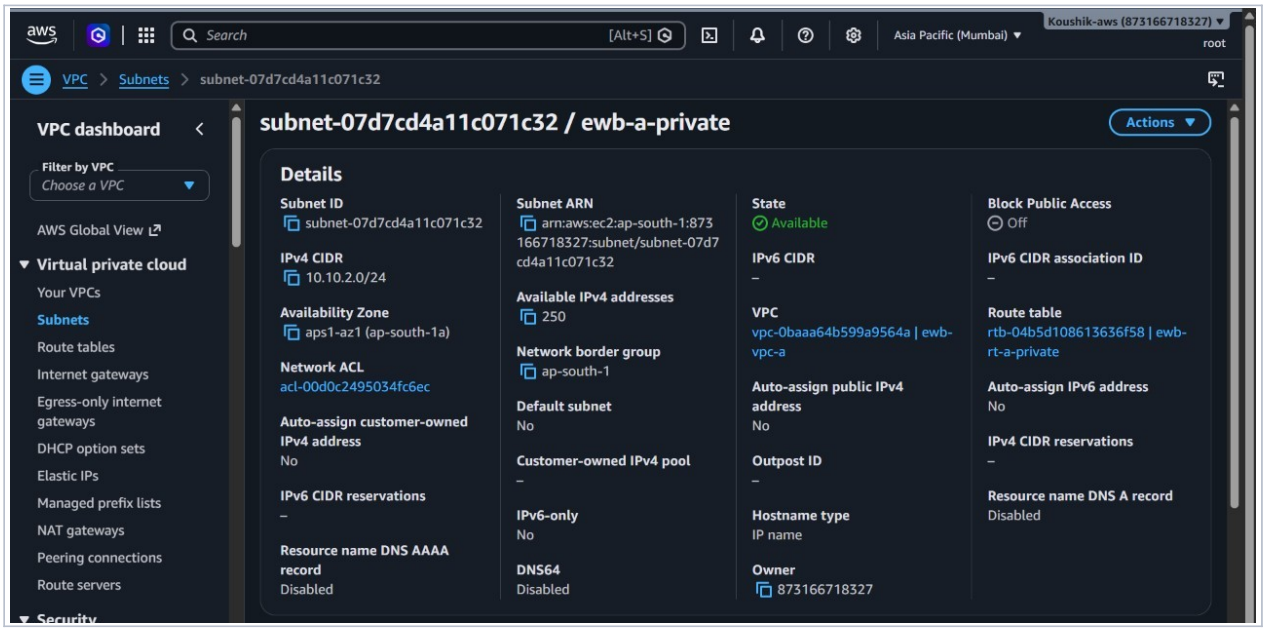


Figure 6. ewb-a-private (subnet-07d7cd4a11c071c32): 10.10.2.0/24, ap-south-1a, auto-assign public IPv4 No, route table ewb-rt-a-private.

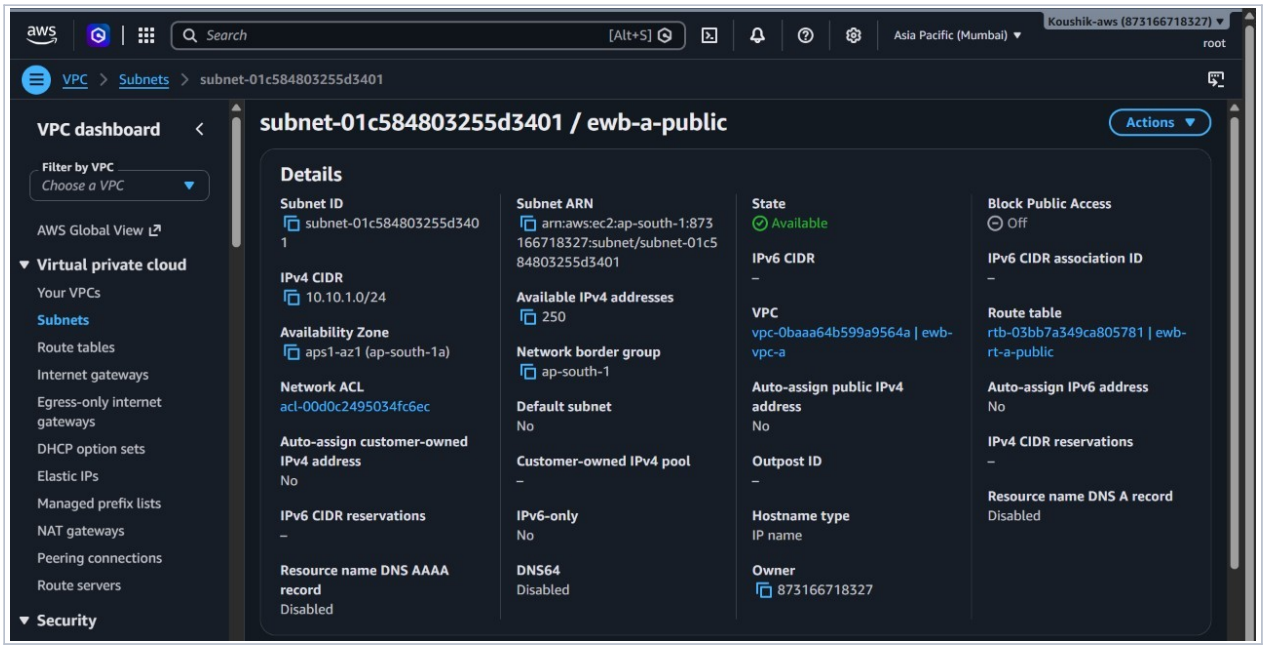


Figure 7. ewb-a-public (subnet-01c584803255d3401): 10.10.1.0/24, ap-south-1a, auto-assign public IPv4 No, route table ewb-rt-a-public.

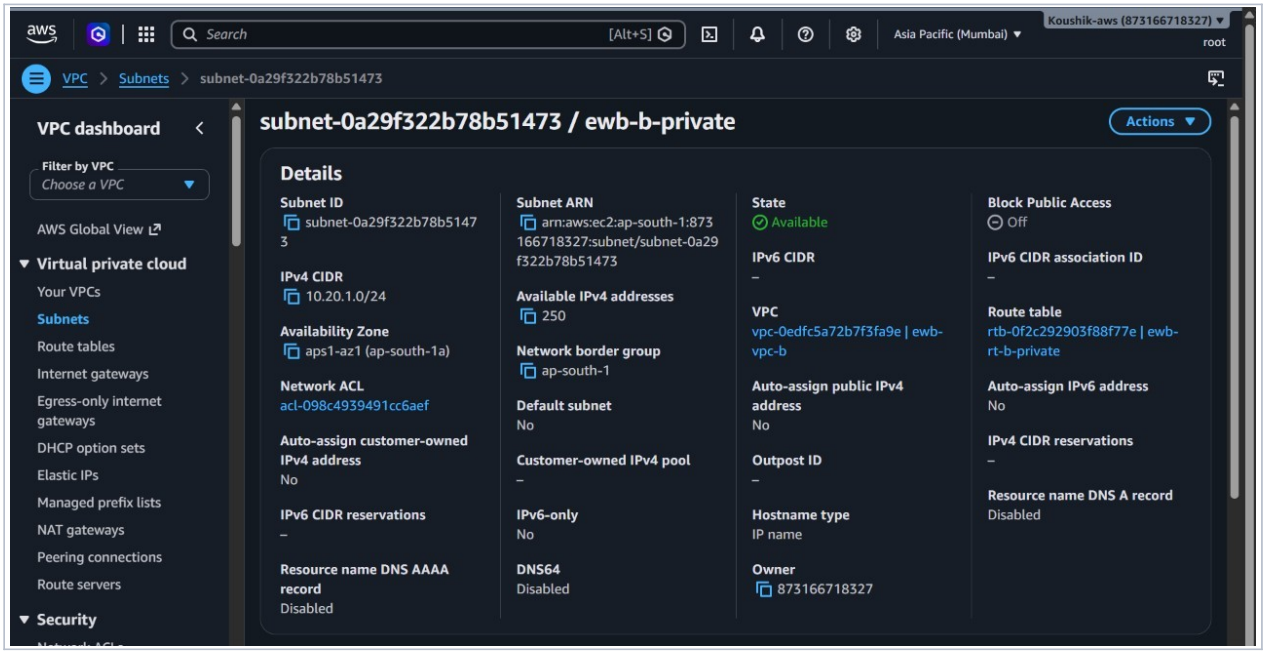


Figure 8. ewb-b-private (subnet-0a29f322b78b51473): 10.20.1.0/24, ap-south-1a, auto-assign public IPv4 No, route table ewb-rt-b-private.

ewb-rt-a-public (VPC A) - rtb-03bb7a349ca805781

Destination	Target	Status	Purpose
0.0.0.0/0	igw-0fafab354c6908f0d	Active	Internet access (SSH to the bastion)
10.10.0.0/16	local	Active	Traffic inside VPC A
10.20.0.0/16	pcx-00d88e5dafce78b7e	Active	Forward route to VPC B through the peering connection

This table is explicitly associated with subnet-01c584803255d3401 (ewb-a-public). Main: No. The target of the 10.20.0.0/16 route is cut off in the screenshot (pcx-00d88e5dafce78b…); the full ID comes from the peering connection screenshot (Figure 13).

ewb-rt-a-private (VPC A) - rtb-04b5d108613636f58

Destination	Target	Status	Purpose
10.10.0.0/16	local	Active	Traffic inside VPC A
10.20.0.0/16	pcx-00d88e5dafce78b7e	Active	Forward route to VPC B through the peering connection

This table is explicitly associated with subnet-07d7cd4a11c071c32 (ewb-a-private), as Figure 12 shows. It has no internet gateway route, so ewb-app-a has no path to the internet.

ewb-rt-b-private (VPC B) - rtb-0f2c292903f88f77e

Destination	Target	Status	Purpose
10.10.0.0/16	pcx-00d88e5dafce78b7e	Active	Return route to VPC A through the peering connection
10.20.0.0/16	local	Active	Traffic inside VPC B

This table is explicitly associated with subnet-0a29f322b78b51473 (ewb-b-private) and also has no internet gateway route.

The route destinations match the VPC CIDRs: 10.20.0.0/16 on the VPC A side and 10.10.0.0/16 on the VPC B side. Because both sides have a route, replies can find their way back. The peering routes show Route Origin ‘Create Route’, and the local routes show ‘Create Route Table’.

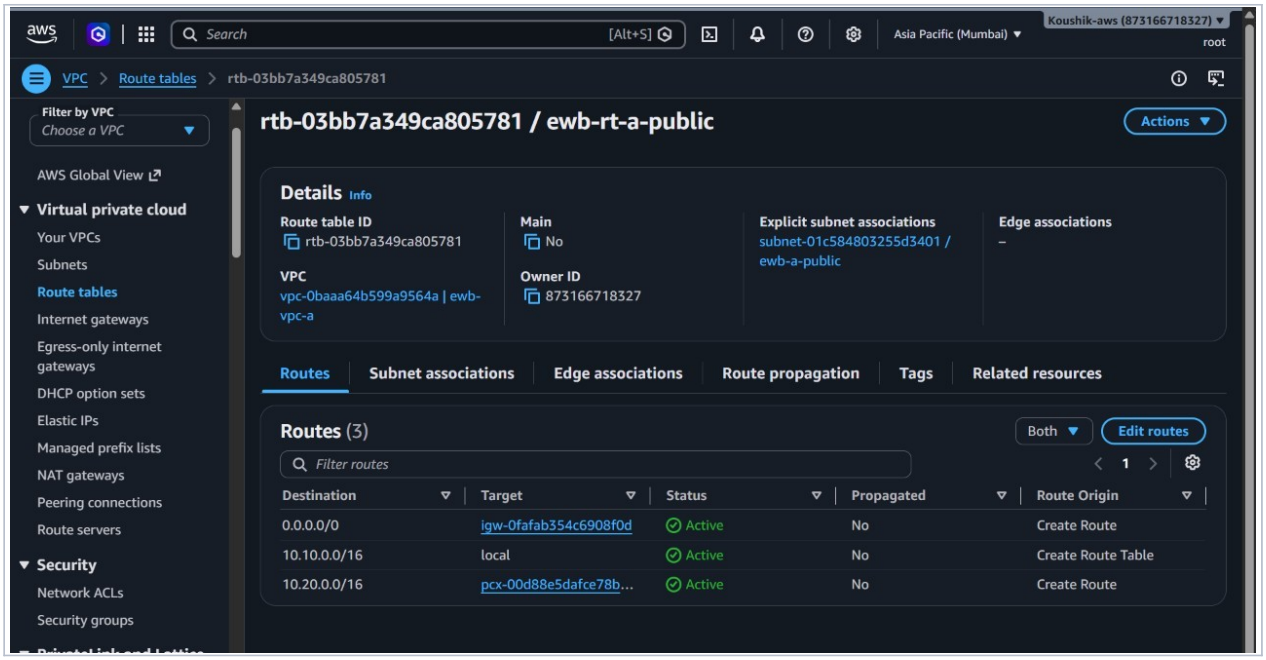


Figure 9. ewb-rt-a-public (rtb-03bb7a349ca805781): three routes, 0.0.0.0/0 to igw-0fafab354c6908f0d, 10.10.0.0/16 local and 10.20.0.0/16 to the peering connection, all Active.

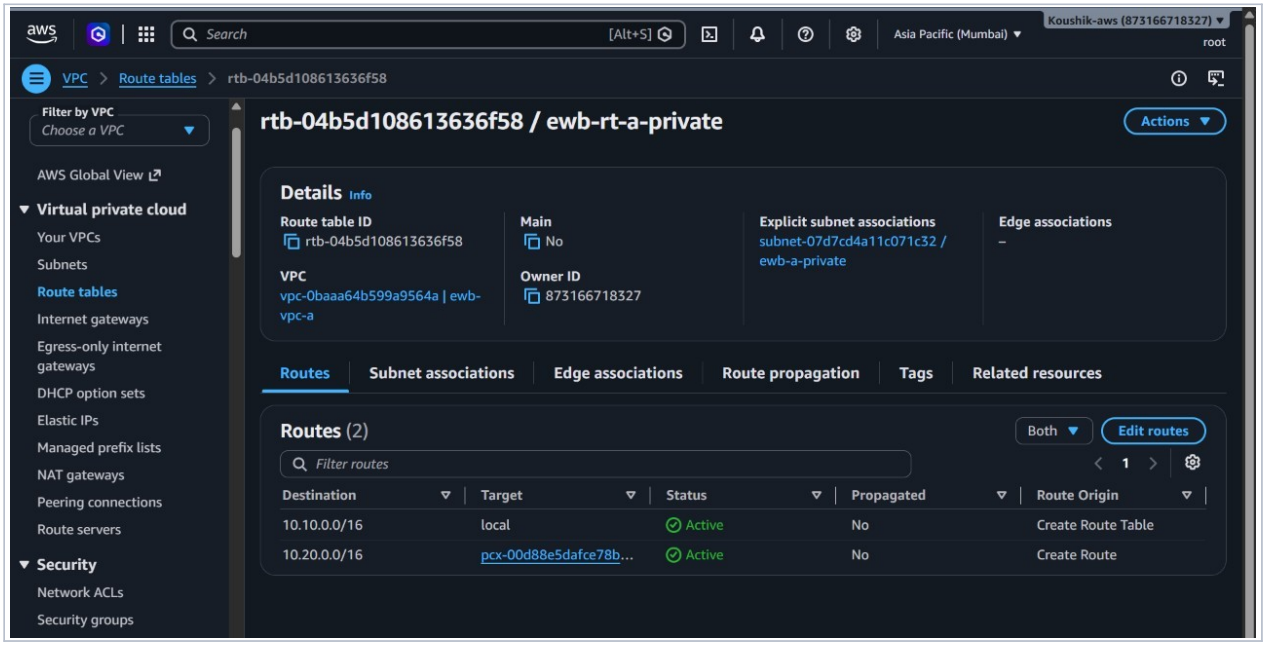


Figure 10. ewb-rt-a-private (rtb-04b5d108613636f58): two routes, 10.10.0.0/16 local and 10.20.0.0/16 to the peering connection, all Active.

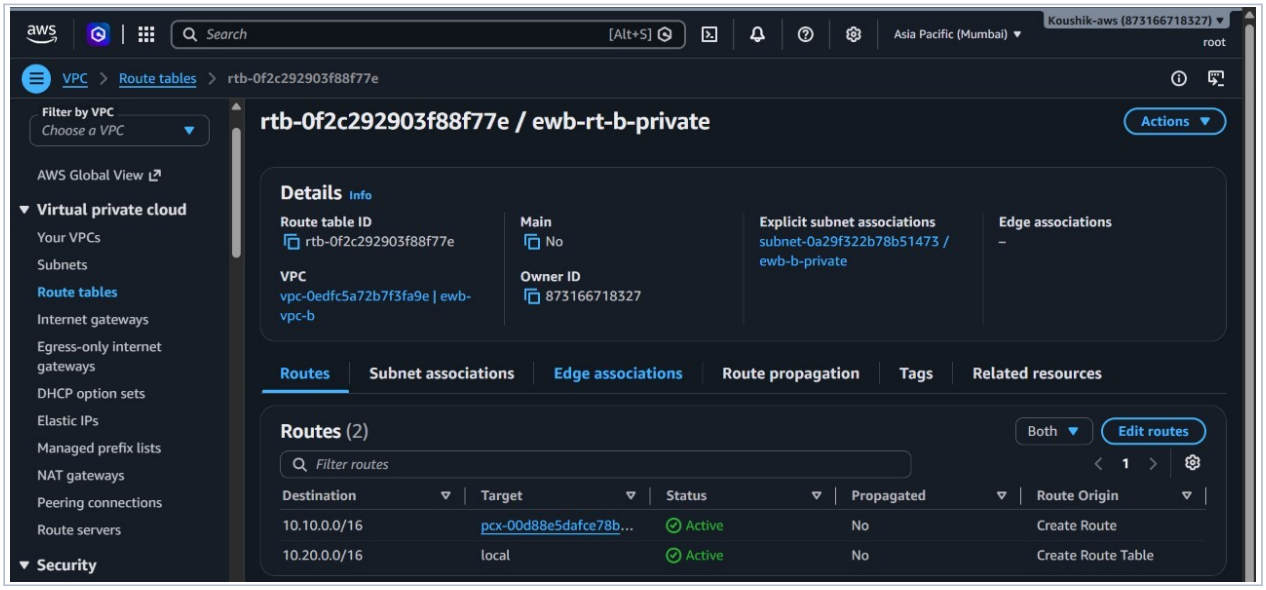


Figure 11. ewb-rt-b-private (rtb-0f2c292903f88f77e): two routes, 10.10.0.0/16 to the peering connection and 10.20.0.0/16 local, all Active.

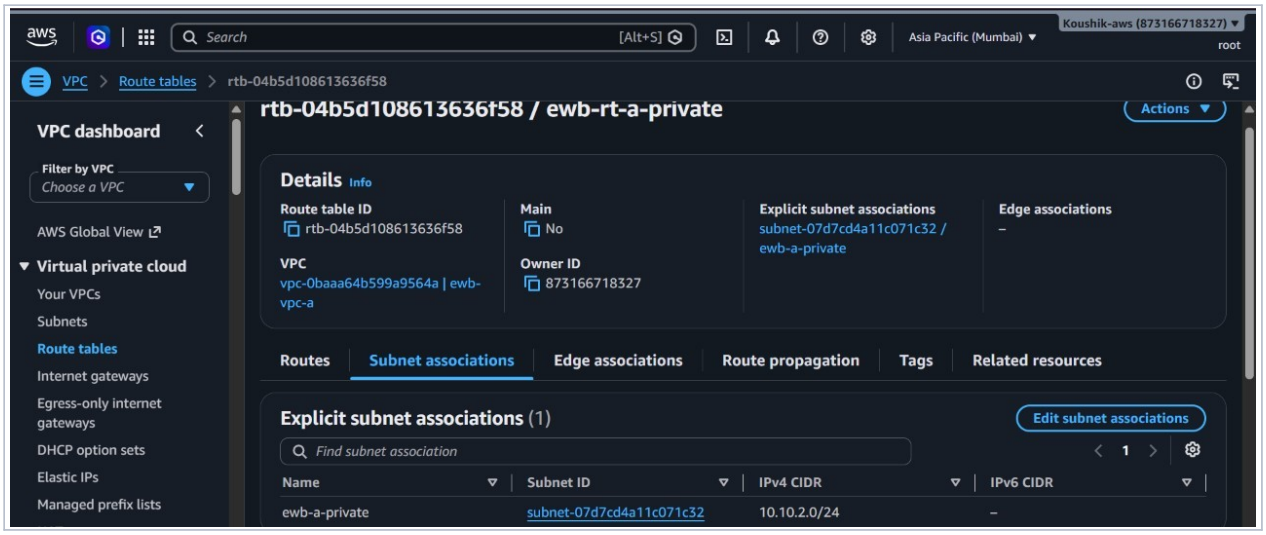


Figure 12. ewb-rt-a-private, Subnet associations tab: one explicit association, ewb-a-private (subnet-07d7cd4a11c071c32, 10.10.2.0/24).

Not captured
I did not capture a route table before the peering routes were added, and the supplied screenshots do not include a route table list or the internet gateway page.

05 / PEERING CONNECTION

Request, accept, activate

The connection is the link. Routes and security groups decide what can use it.

Field	Value (from Figures 13 and 25)
Name / ID	ewb-a-b-peer / pcx-00d88e5dafce78b7e
Status	Active

Field	Value (from Figures 13 and 25)
Requester VPC	vpc-0baaa64b599a9564a (ewb-vpc-a)
Accepter VPC	vpc-0edfc5a72b7f3fa9e (ewb-vpc-b)
Requester and accepter owner	Same AWS account (both owner IDs are the same)
Peering connection ARN	arn:aws:ec2:ap-south-1:…:vpc-peering-connection/pcx-00d88e5dafce78b7e (ap-south-1)
Requester and accepter Region	Mumbai (ap-south-1) for both (Figure 25)
CIDRs and DNS settings	Not captured: the CIDRs show '–' in the later screenshot (Figure 25) and the DNS settings are below the visible part of Figure 13

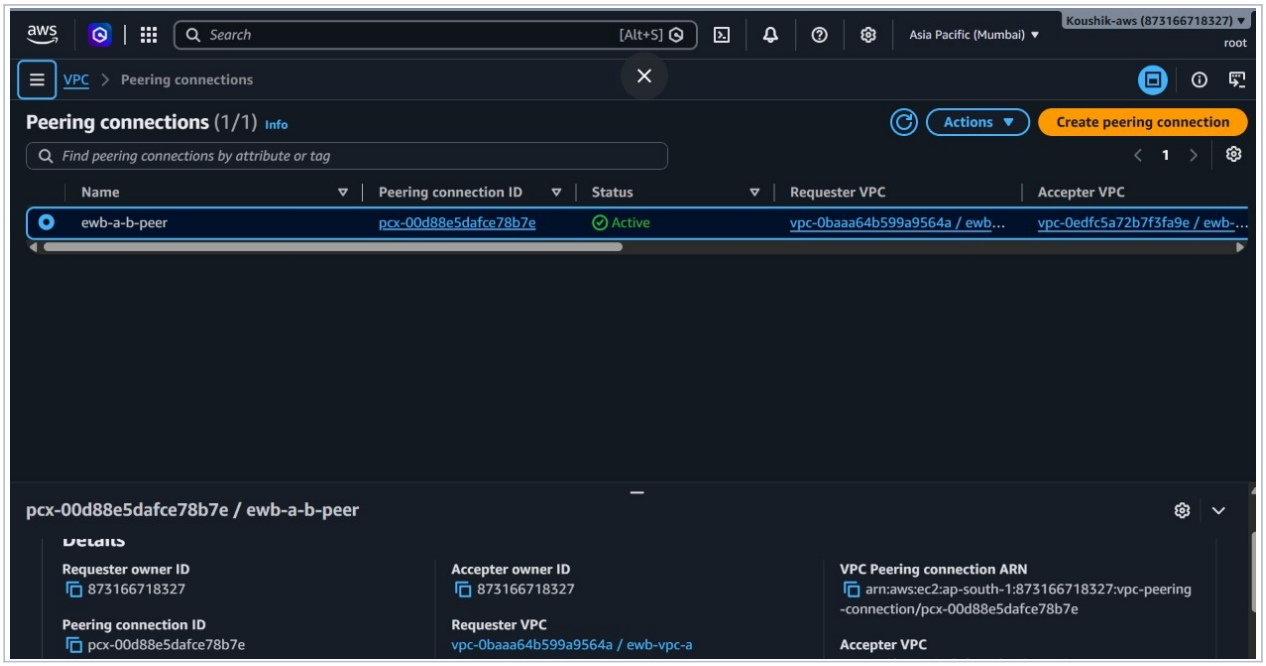


Figure 13. Peering connection pcx-00d88e5dafce78b7e / ewb-a-b-peer: Status Active, requester vpc-0baaa64b599a9564a (ewb-vpc-a), accepter vpc-0edfc5a72b7f3fa9e, same owner account.

06 / SECURITY GROUPS

Inbound rules

Only the traffic needed for SSH and ping.

Security group 1 (bastion) - sg-0ef812dc16c705a9d

Rule ID	Type	Protocol / port	Source
sgr-0af1995d17bcbfac1	SSH	TCP / 22	157.50.125.69/32 (a single public address)

Security group 2 (ewb-app-a) - sg-0e267dfedbcba142f

Rule ID	Type	Protocol / port	Source
sgr-0e83d10439c59f552	All ICMP - IPv4	ICMP / All	10.20.1.10/32 (ewb-app-b)
sgr-020f39df5f2bb94db	SSH	TCP / 22	10.10.1.10/32 (ewb-bastion)

Security group 3 (ewb-app-b) - sg-02ab0878c5cbfeaf9

Rule ID	Type	Protocol / port	Source
sgr-0909691efad8a5706	All ICMP - IPv4	ICMP / All	10.10.2.10/32 (ewb-app-a)
sgr-0580a35333510d54b	SSH	TCP / 22	10.10.1.10/32 (ewb-bastion)

Each rule uses a single /32 address. SSH to the bastion is open to one public address only, SSH to the two app instances is allowed only from the bastion's private IP, and ICMP is allowed only from the other app instance. ICMP from the peer's /32 matches the workbook's single-address Echo rule, except that the type is 'All ICMP - IPv4' rather than Echo request only.

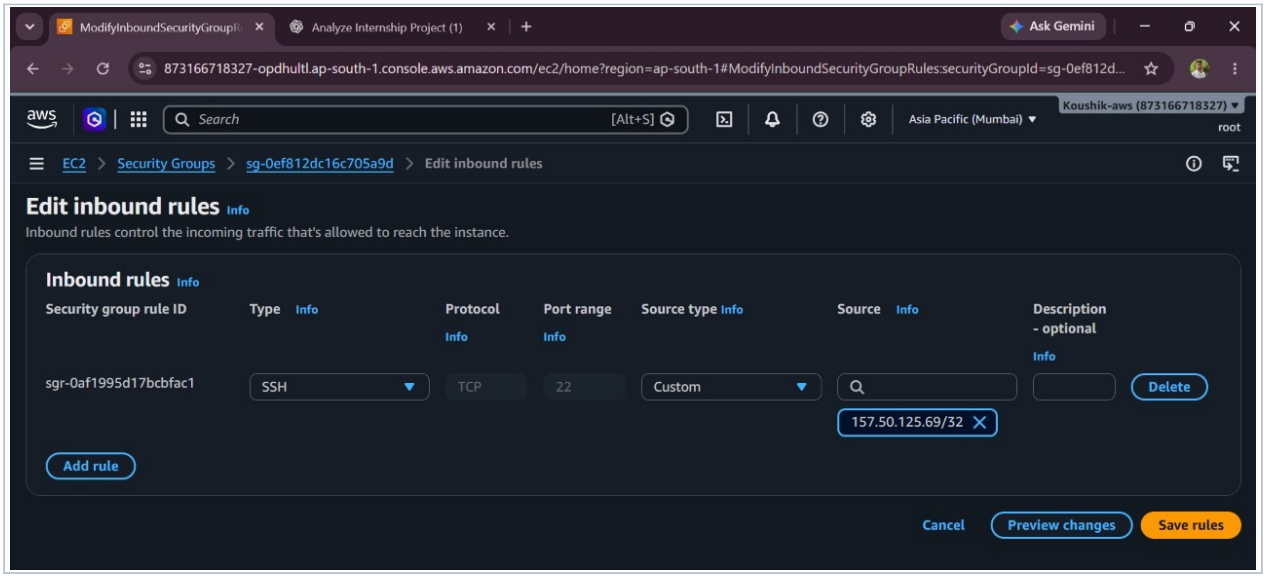


Figure 14. Edit inbound rules for sg-0ef812dc16c705a9d: SSH (TCP 22) from 157.50.125.69/32.

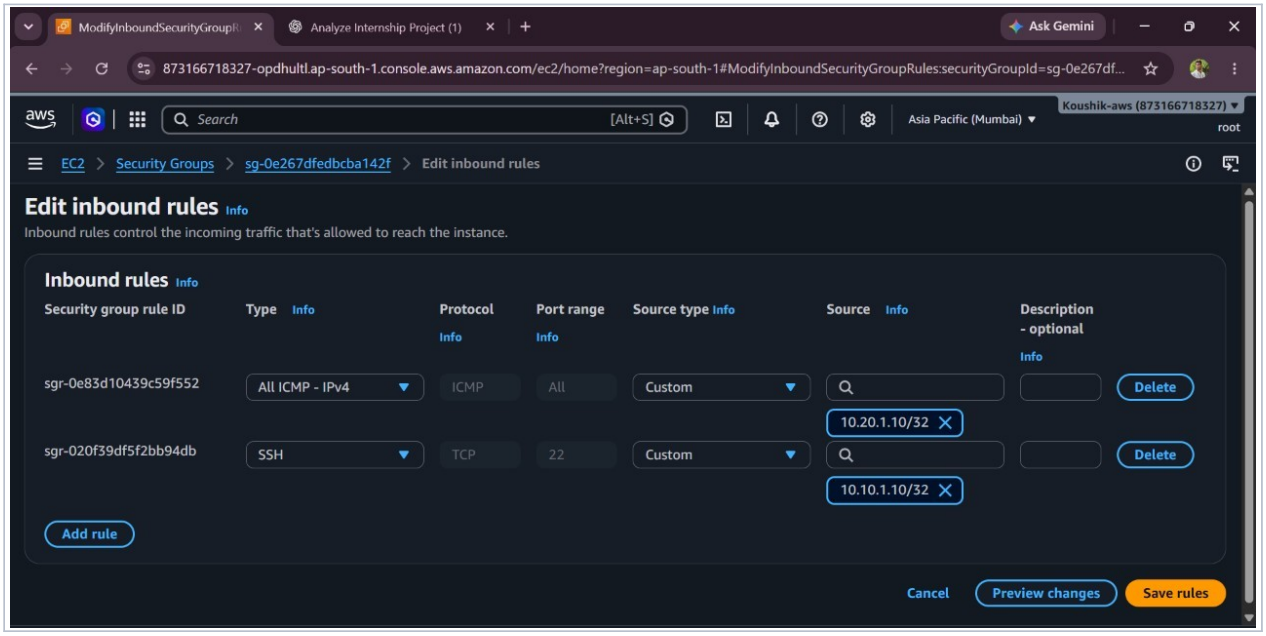


Figure 15. Edit inbound rules for sg-0e267dfedbcba142f: All ICMP - IPv4 from 10.20.1.10/32 and SSH from 10.10.1.10/32.

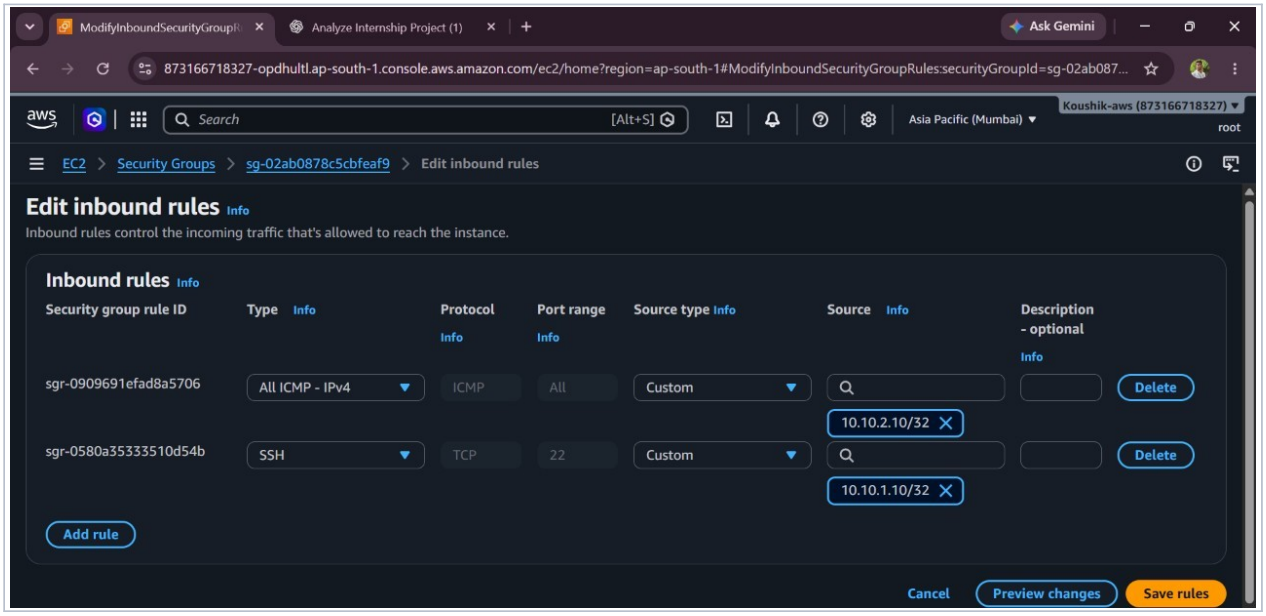


Figure 16. Edit inbound rules for sg-02ab0878c5cbfeaf9: All ICMP - IPv4 from 10.10.2.10/32 and SSH from 10.10.1.10/32.

07 / COMPUTE

EC2 instances

Three Amazon Linux 2023 instances. Only the bastion has a public IP.

Field	ewb-bastion	ewb-app-a	ewb-app-b
Instance ID	i-07a957f5791d232bc	i-09a5f9baa26eaafb8	i-01695738463794277
State	Running	Running	Running
Type	t3.micro	t3.micro	t3.micro
VPC	ewb-vpc-a	ewb-vpc-a	ewb-vpc-b
Subnet	ewb-a-public	ewb-a-private	ewb-b-private
Private IPv4	10.10.1.10	10.10.2.10	10.20.1.10
Public IPv4	43.205.111.8 (auto-assigned)	None (field empty)	None (field empty)
Elastic IP	None	None	None
IAM role	EWB-Bastion-SSM-Role	None	None
IMDSv2	Required	Required	Required
Security group	Not captured	Not captured	Not captured

The Public IPv4 field for ewb-app-a and ewb-app-b is empty, so neither instance has a direct internet address. The only way in from my laptop is through the bastion. The security group of each instance is below the visible part of the screenshots, so I have marked it ‘Not captured’.

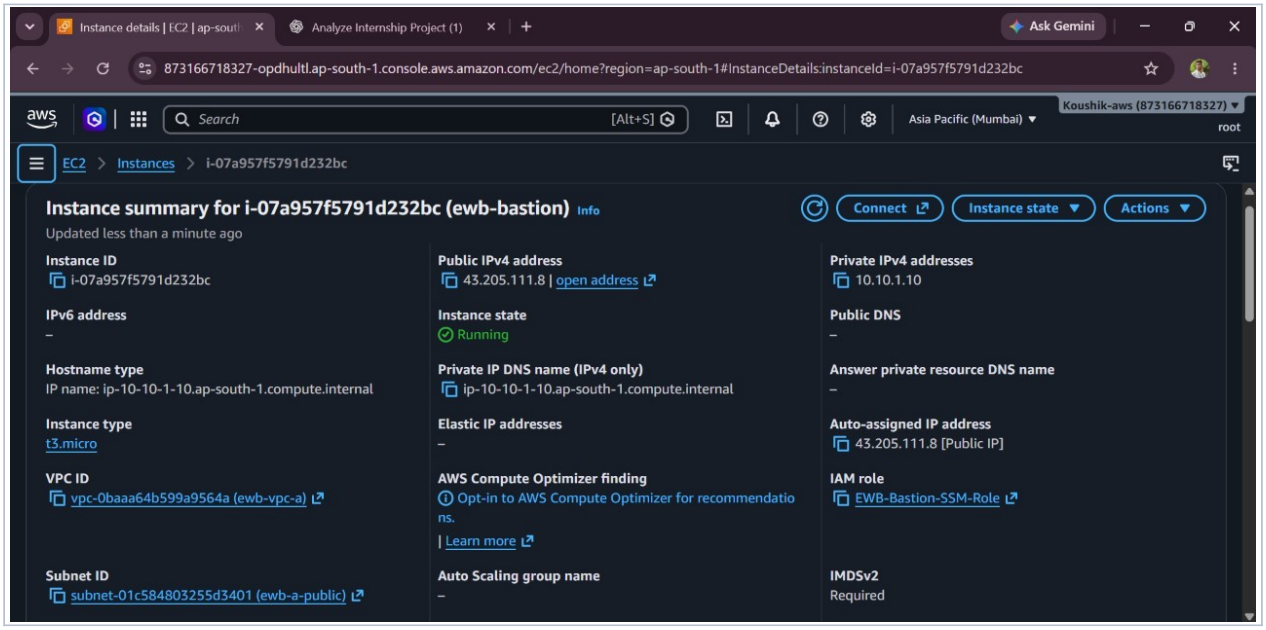


Figure 17. ewb-bastion (i-07a957f5791d232bc): Running, t3.micro, private IPv4 10.10.1.10, public IPv4 43.205.111.8, IAM role EWB-Bastion-SSM-Role, subnet ewb-a-public.

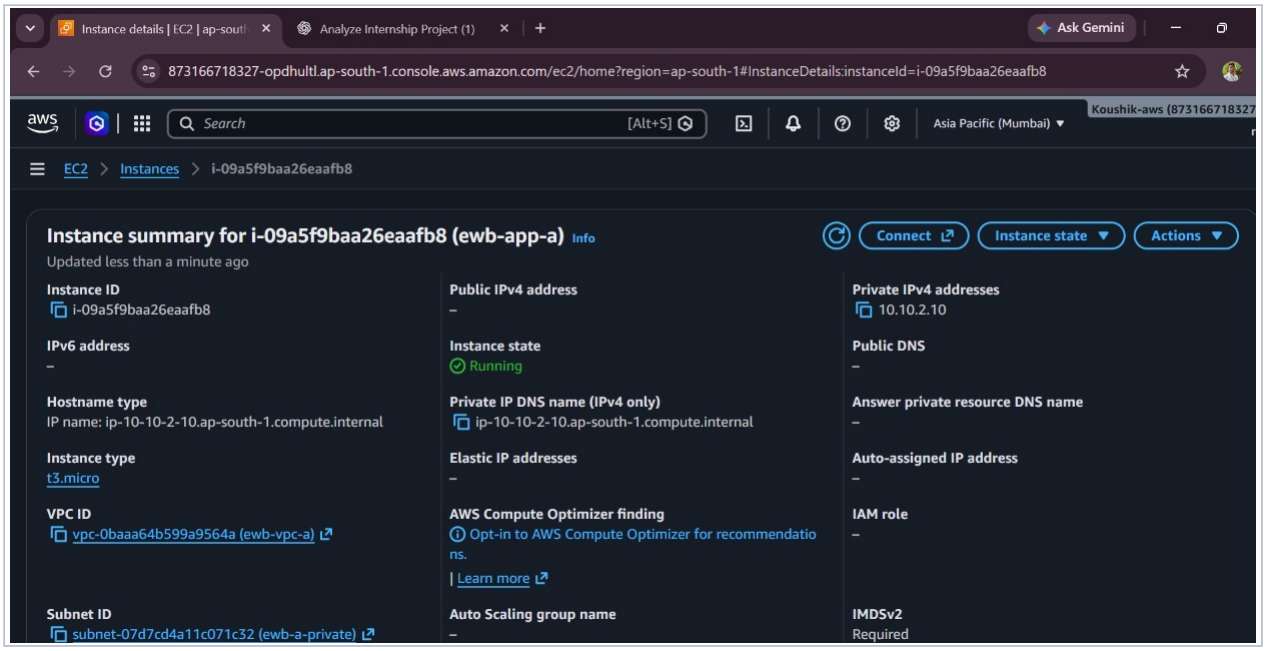


Figure 18. ewb-app-a (i-09a5f9baa26eaafb8): Running, t3.micro, private IPv4 10.10.2.10, no public IPv4, subnet ewb-a-private.

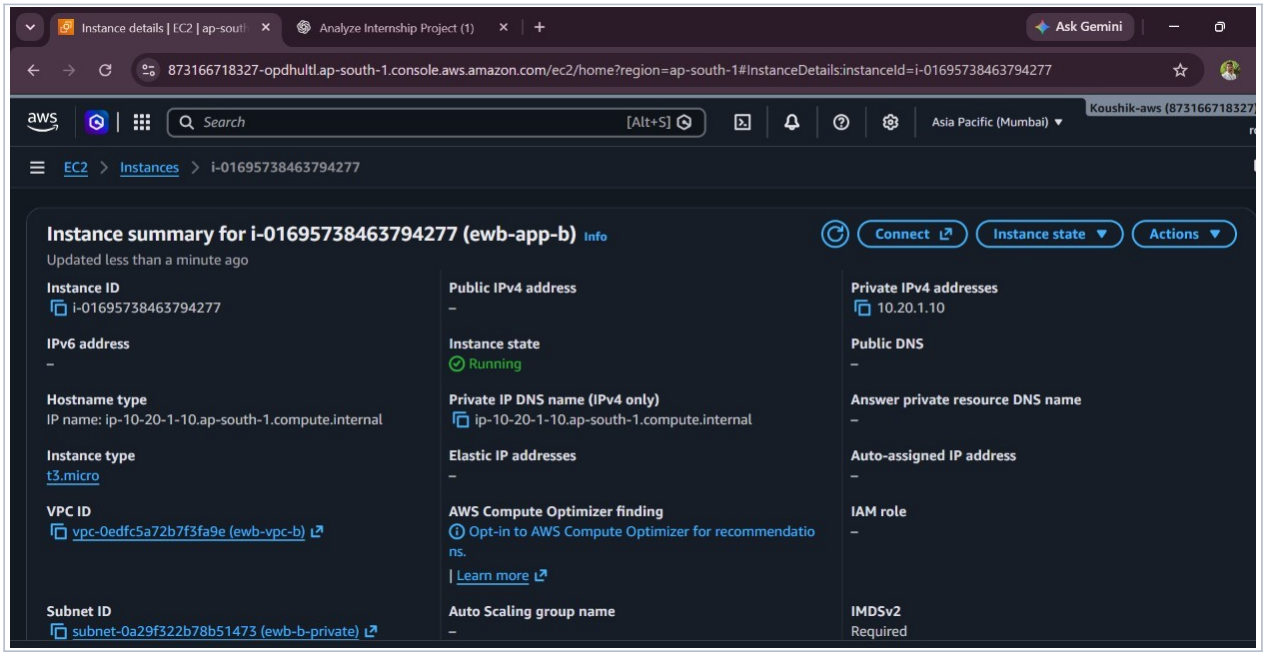


Figure 19. ewb-app-b (i-01695738463794277): Running, t3.micro, private IPv4 10.20.1.10, no public IPv4, VPC ewb-vpc-b, subnet ewb-b-private.

08 / VERIFICATION

Connectivity tests

Results from my terminal screenshots.

Test	Where / command	Result	Figure
Key in SSH agent	Laptop: ssh-add -l	One RSA 2048 key listed (ewb-lab.pem)	Fig 20
SSH to ewb-app-a via the bastion	Laptop: ssh -A -o IdentitiesOnly=no -J ec2-user@43.205.111.8 ec2-user@10.10.2.10	Host key accepted; login worked (Amazon Linux 2023), prompt ip-10-10-2-10	Fig 20
Forward ping	ewb-app-a: ping -c 4 10.20.1.10	4 sent, 4 received, 0% loss, time 3111 ms; rtt min/avg/max/mdev 0.169/0.263/0.482/0.127 ms	Fig 21
Isolation	EC2 instance details	ewb-app-a and ewb-app-b: Public IPv4 address is empty	Fig 18, 19

In Figure 20 I used ssh with -J (ProxyJump) so that the connection to ewb-app-a goes through the bastion's public IP, 43.205.111.8, and -A so that my key can be used from the agent for the second hop. The login banner and the prompt ip-10-10-2-10 confirm that I reached ewb-app-a. In Figure 21 the replies come from 10.20.1.10, the private IP of ewb-app-b, with TTL 127.

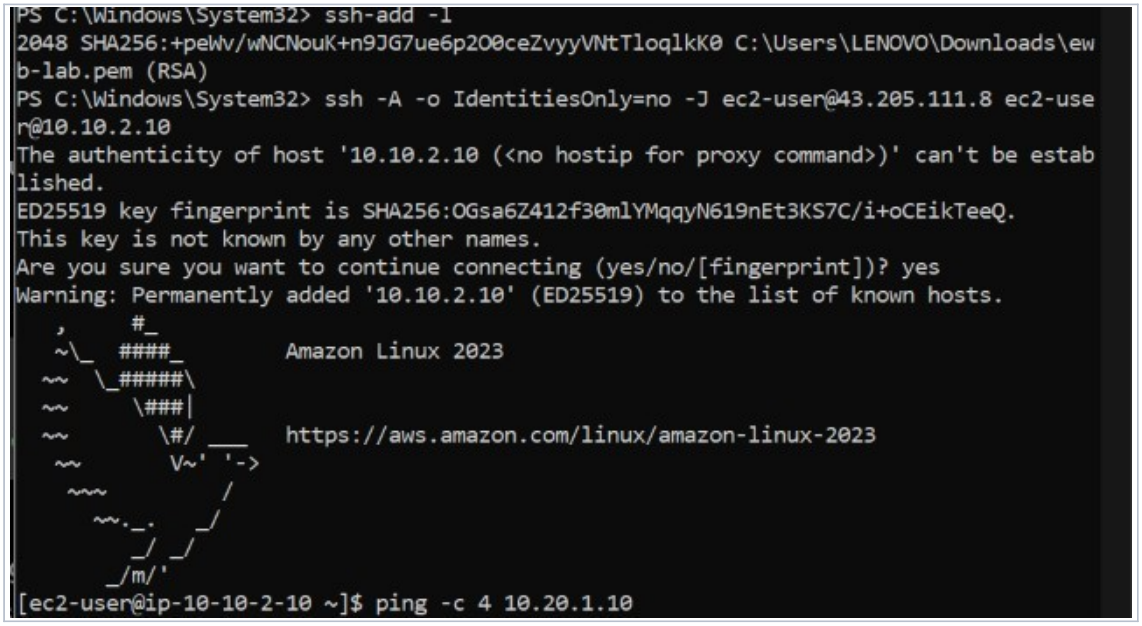


Figure 20. Laptop PowerShell: ssh-add -l, then ssh -A -J through 43.205.111.8 to ec2-user@10.10.2.10 (ewb-app-a). The prompt shows ip-10-10-2-10. The last line starts ping -c 4 10.20.1.10.

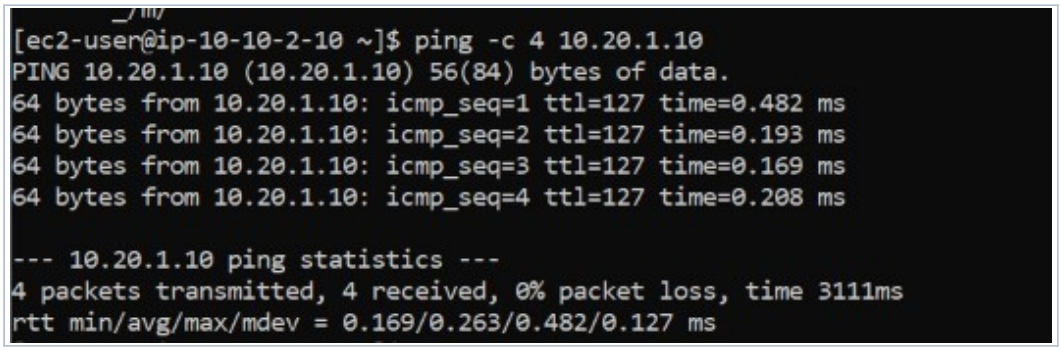


Figure 21. From ewb-app-a: ping -c 4 10.20.1.10 gave 4 sent, 4 received, 0% packet loss (rtt min/avg/max 0.169/0.263/0.482 ms).

Not captured
The reverse ping from ewb-app-b, the SSH login to ewb-app-b and a ping or SSH test from before peering are not in my screenshots.

09 / FAILURE AND RECOVERY

ICMP failure and recovery

What my screenshots show and what they do not.

Step	Command (from ewb-app-a)	Result	Figure
Failure	Not captured	Not captured	–
Ping to ewb-app-b	ping -c 4 10.20.1.10	4 sent, 4 received, 0% loss; rtt min/avg/max 0.169/0.263/0.482 ms	Fig 20, 21

What I cannot show
In the workbook, the controlled failure is done by removing the destination's inbound ICMP rule and adding it back. I do not have a screenshot of a failed ping or of a rule being removed or restored, so I cannot show a failure, say what caused one, or show a recovery. Figure 20 below is the SSH session on ewb-app-a where the ping command was started; the screenshot ends at the command, with no result visible. The result of that ping is in Figure 21 (0% packet loss).

In general, when ping fails while the peering connection is Active, the problem is usually a route, a security group, a network ACL or the instance firewall, not the peering connection itself. That is general VPC behaviour. My screenshots do not test it for this case.

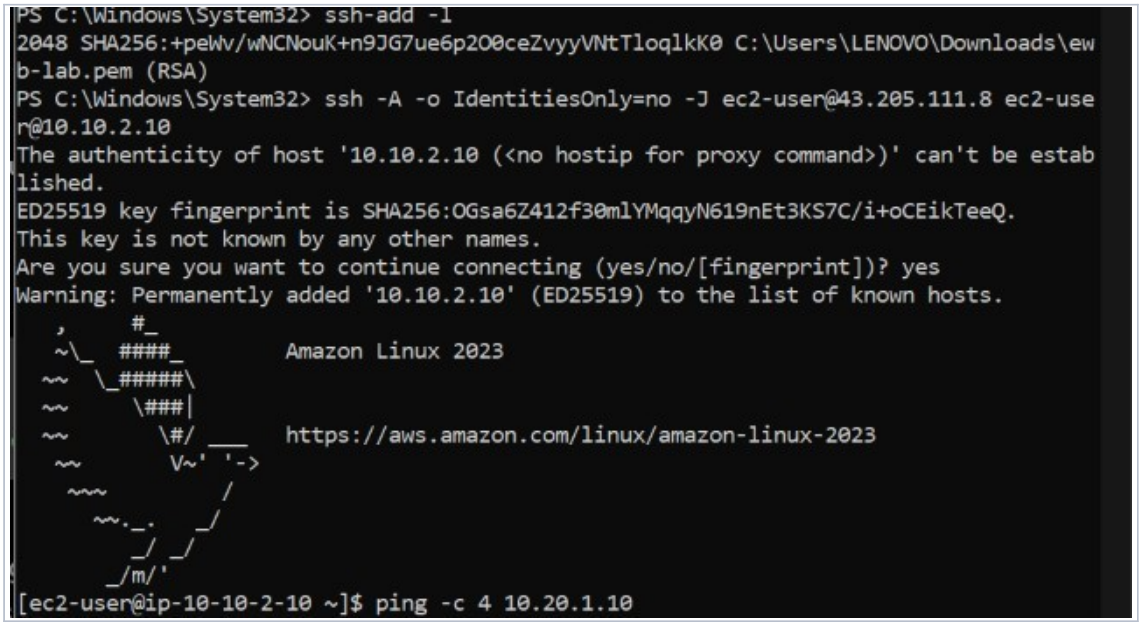


Figure 20 (repeated from Section 8). Laptop PowerShell: ssh through the bastion to ewb-app-a (prompt ip-10-10-2-10), then ping -c 4 10.20.1.10 is started. The screenshot ends at the command.

10 / TRAFFIC PATHS

Forward path and return path

Why both routes and both security rules are needed.

Forward path: ewb-app-a (10.10.2.10) to ewb-app-b (10.20.1.10)

1. The destination 10.20.1.10 matches 10.20.0.0/16 in ewb-rt-a-private, whose target is pcx-00d88e5dafce78b7e. There is no internet gateway route in this table.
2. The peering connection delivers the packet into VPC B.
3. The security group of ewb-app-b has to allow it: its ICMP rule allows 10.10.2.10/32, which is ewb-app-a. Figure 21 shows the ping working.

Return path: ewb-app-b back to ewb-app-a

1. The reply's destination 10.10.2.10 matches 10.10.0.0/16 in ewb-rt-b-private, which points to the same peering connection.
2. Security groups are stateful, so replies to allowed traffic are allowed automatically.

3. A ping started from ewb-app-b is also allowed by the configuration: the ewb-app-a group allows ICMP from 10.20.1.10/32. I did not capture a test of it.

Administration path

My laptop connects to 43.205.111.8 on TCP 22 through the internet gateway, using the 0.0.0.0/0 route in ewb-rt-a-public and the SSH rule for my public address on the bastion's security group. From ewb-bastion I then connect to 10.10.2.10 inside VPC A. The ewb-app-a group allows SSH only from 10.10.1.10/32. ewb-app-a and ewb-app-b have no public IPv4 address and no internet gateway route, so the bastion is the only way in from my laptop.

11 / CLEANUP

Cleanup and final resource state

Everything I deleted and what the screenshots show.

I have 11 cleanup screenshots. The two Instances screenshots keep the filter 'Instance state = running' (Figures 22 and 23); the other lists show no filter. The screenshots do not show a time, so I do not give one.

Workbook cleanup step	Status	Figure
Terminate ewb-bastion, ewb-app-a and ewb-app-b	Done: all three instances show Terminated, and the running-instances filter shows no matching instances	Fig 22, 23
Release Elastic IPs	Done: no Elastic IP addresses in the Region	Fig 24
Remove unused lab key pairs	Not captured	–
Delete peering connection and peer routes	Done: status Deleted, and ewb-rt-a-public no longer has the 10.20.0.0/16 route	Fig 25, 26
Delete subnets	Done: only the three default-VPC subnets (172.31.x.x) are listed	Fig 27
Delete security groups	Done: only the default security group is listed	Fig 28
Delete custom route tables	Done: only one route table is listed, a main route table	Fig 29
Detach and delete the internet gateway	Done: the project gateway is not listed; only igw-0c57f103f7de3e92d, attached to the default VPC, is left	Fig 30
Delete both VPCs	Done: only the default VPC is listed	Fig 31
Check Billing for remaining resources	Month-to-date cost is $11.91 and the forecast says 'Data unavailable'. The page does not show what the cost is for.	Fig 32

Things to note

Figure 26 shows ewb-rt-a-public with two routes, 0.0.0.0/0 to igw-0fafab354c6908f0d and 10.10.0.0/16 local. The route to 10.20.0.0/16 through the peering connection is no longer listed. I did not capture the other two project route tables after the peering was removed; they are simply missing from the Route tables list in Figure 29.

I did not capture volumes, snapshots or key pairs, so I cannot show their state.

Figure 32 shows a month-to-date cost of $11.91, last month's cost for the same period (Sep 1–22) and last month's total both at $0.00, Budgets status 'Setup required' (no budget created), and 2 cost anomalies detected with a total impact of $1.16 over the last 90 days. The page does not show which services the $11.91 comes from, so I cannot say whether the lab caused it. Billing data can also lag, so I cannot tell from this page whether anything is still billing.

Final resource state

In my last screenshots for ap-south-1 the three instances show Terminated and none are running, there are no Elastic IP addresses, and the peering connection shows Deleted. The only VPC, subnets, security group and internet gateway listed belong to the default VPC (172.31.x.x subnets, the default security group sg-0d26796701fecad03, and igw-0c57f103f7de3e92d attached to vpc-084a7dc8cd5639b2a). The only route table listed is a main route table (the VPC column is cut off). None of my project's VPCs, subnets, route tables, internet gateway or security groups is listed.

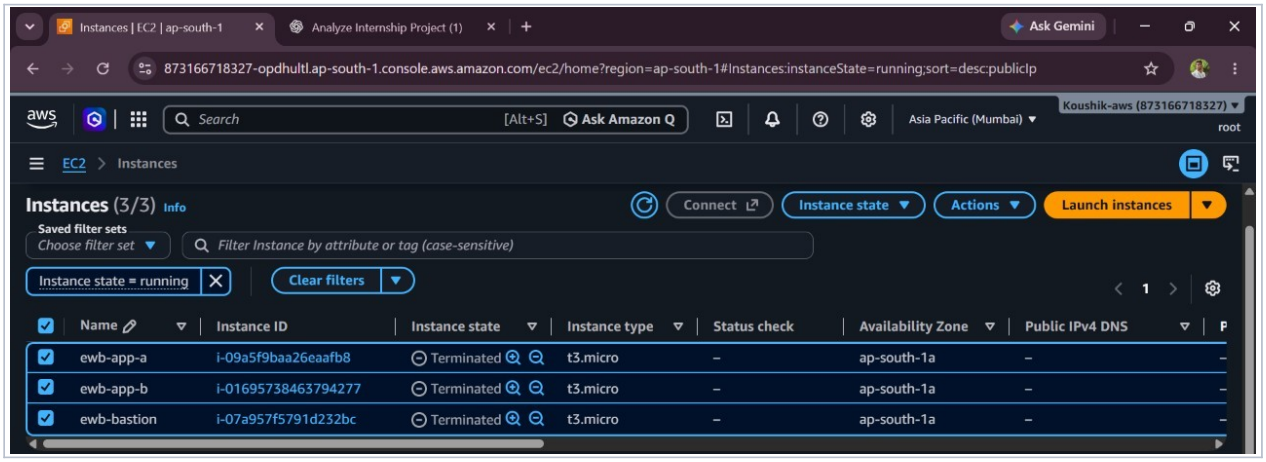


Figure 22. Cleanup, instances: ewb-app-a (i-09a5f9baa26eaafb8), ewb-app-b (i-01695738463794277) and ewb-bastion (i-07a957f5791d232bc) all show Terminated (t3.micro, ap-south-1a).

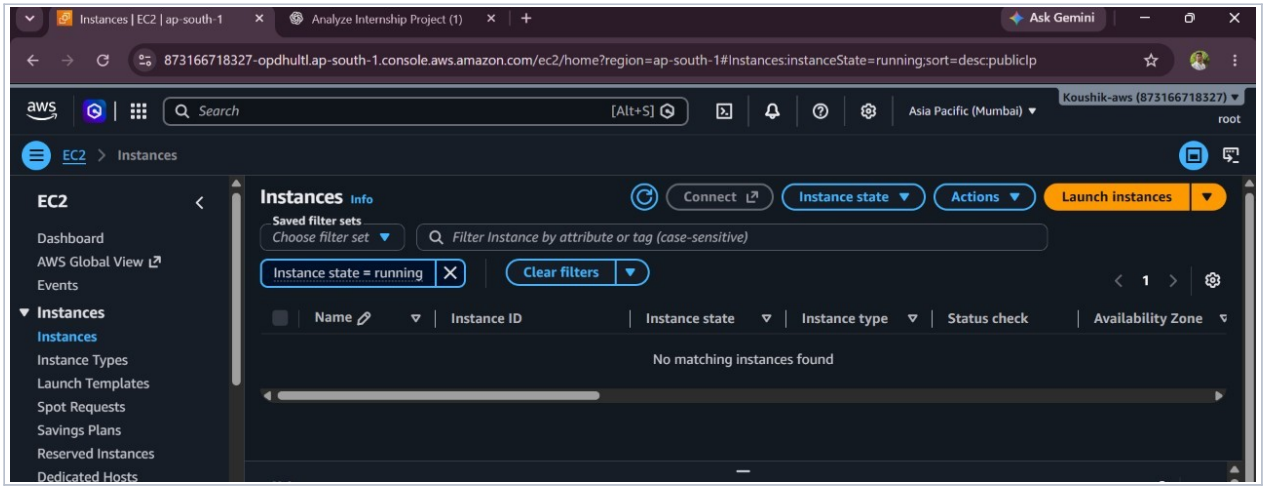


Figure 23. Cleanup, instances: with the filter Instance state = running, the list says 'No matching instances found'.

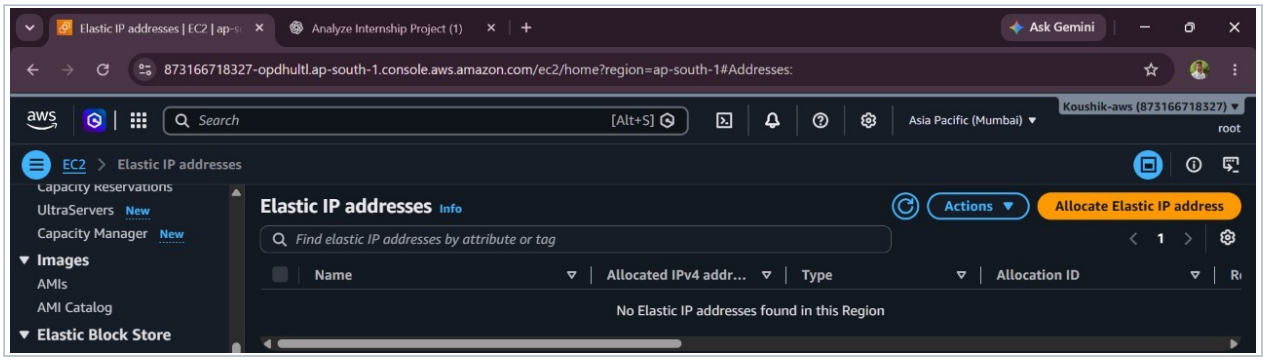


Figure 24. Cleanup, Elastic IPs: 'No Elastic IP addresses found in this Region'.

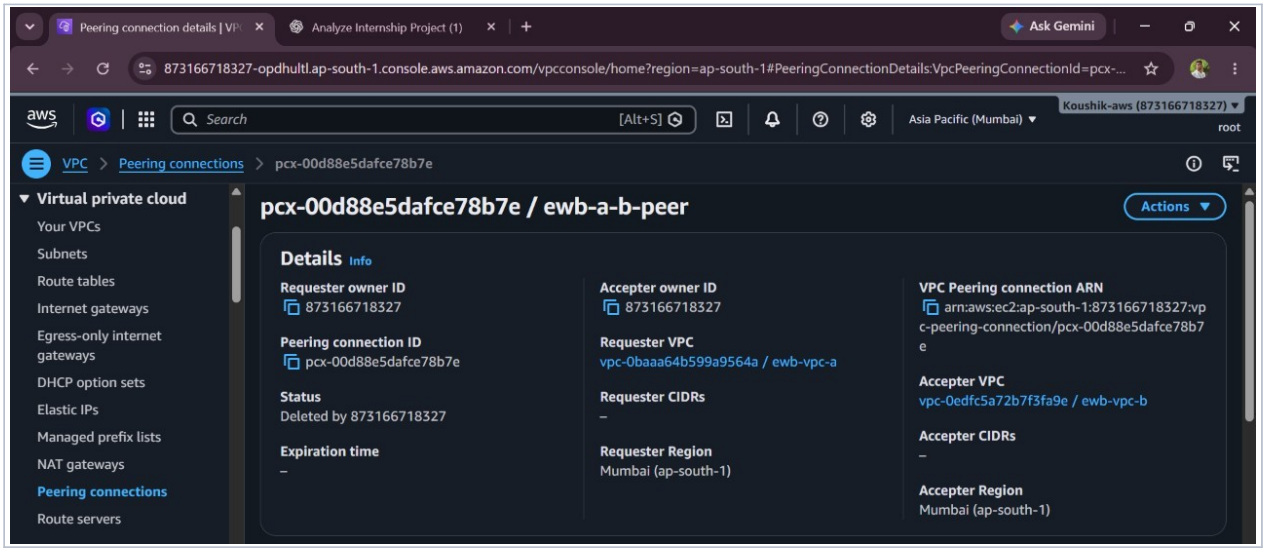


Figure 25. Cleanup, peering: pcx-00d88e5dafce78b7e / ewb-a-b-peer shows Status Deleted. Requester vpc-0baaa64b599a9564a (ewb-vpc-a), accepter vpc-0edfc5a72b7f3fa9e (ewb-vpc-b), both in Mumbai (ap-south-1).

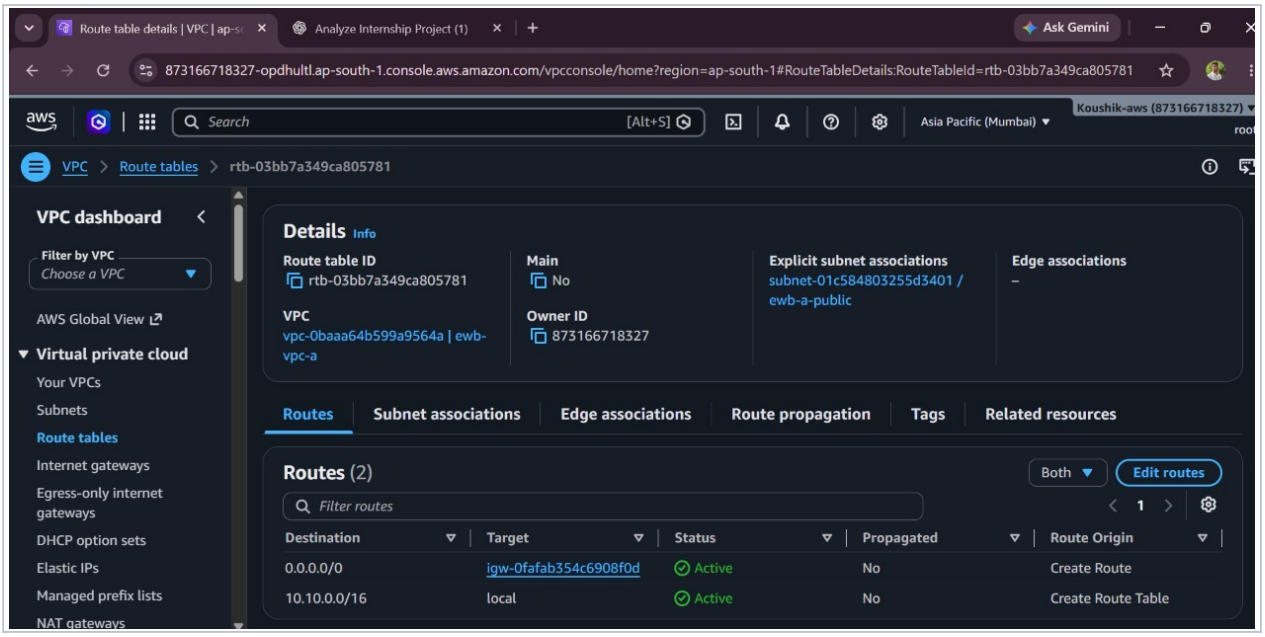


Figure 26. Cleanup, routes: ewb-rt-a-public (rtb-03bb7a349ca805781) now has two routes, 0.0.0.0/0 to igw-0fafab354c6908f0d and 10.10.0.0/16 local. The peering route is gone.

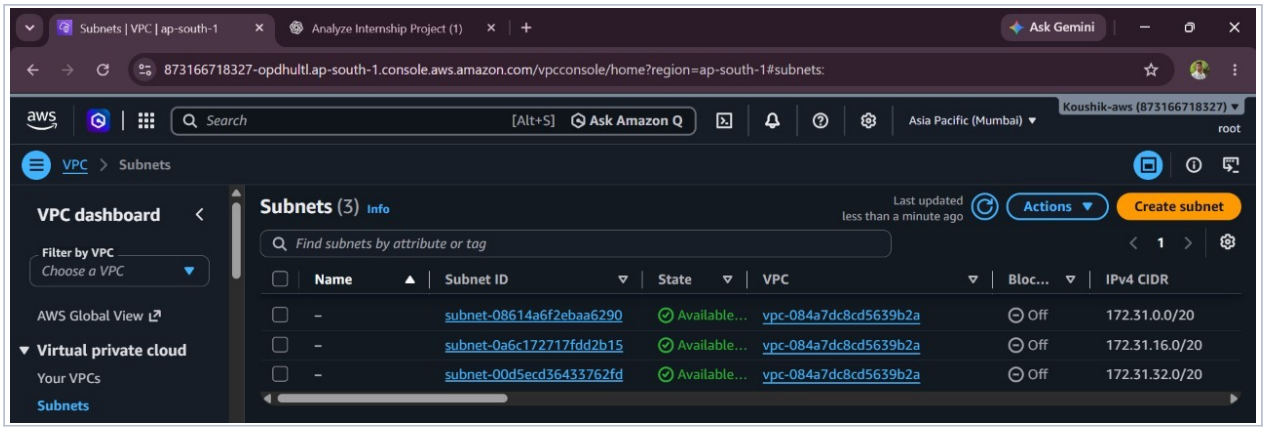


Figure 27. Cleanup, subnets: three subnets are listed, all in the default VPC (172.31.0.0/20, 172.31.16.0/20 and 172.31.32.0/20).

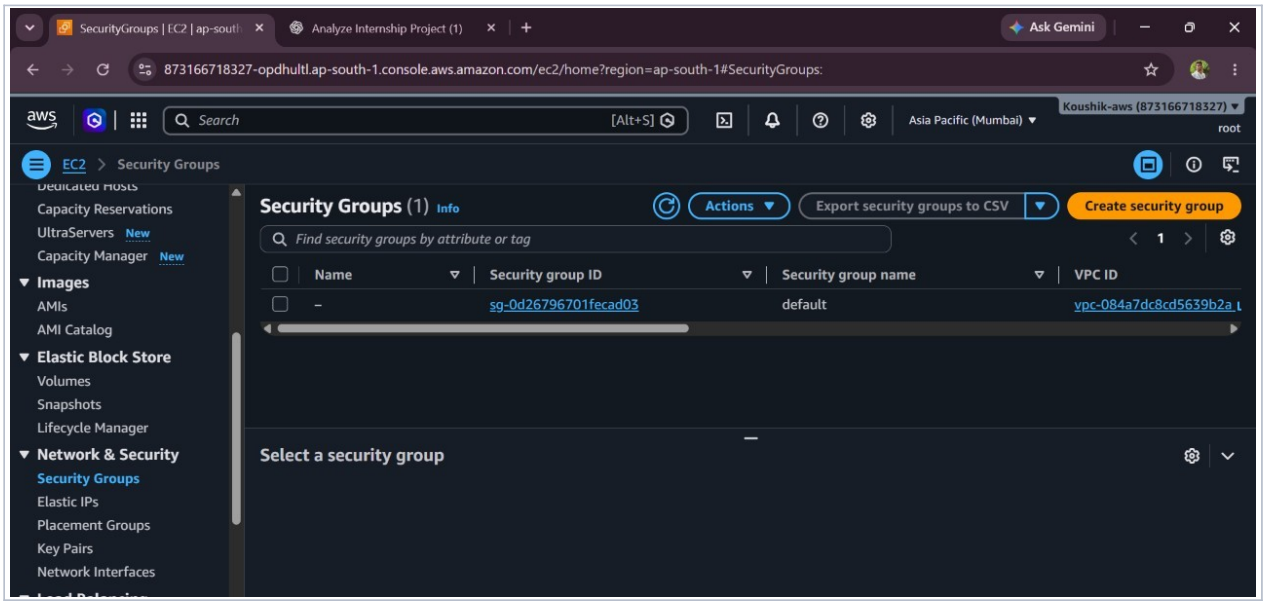


Figure 28. Cleanup, security groups: only the default group sg-0d26796701fecad03 is listed (VPC vpc-084a7dc8cd5639b2a).

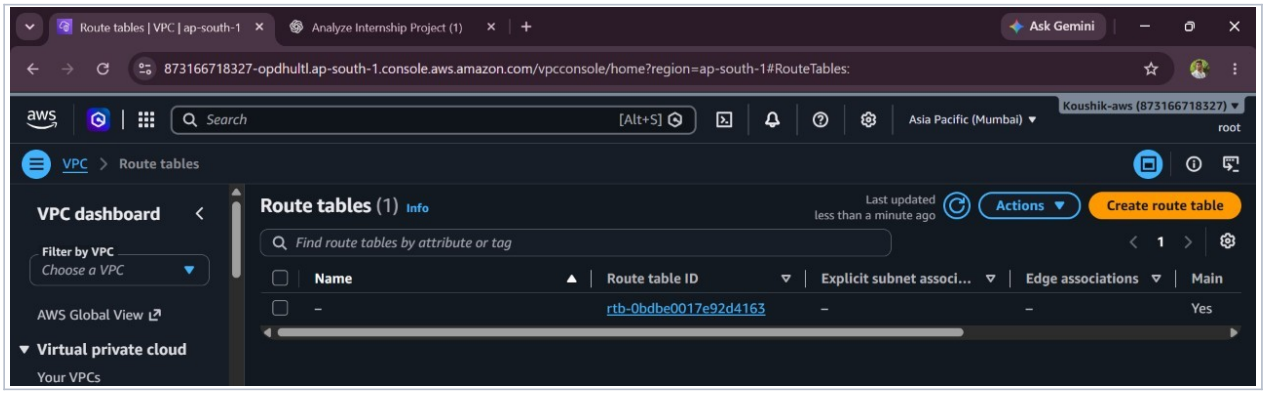


Figure 29. Cleanup, route tables: one route table is listed, rtb-0bdbe0017e92d4163 (Main: Yes).

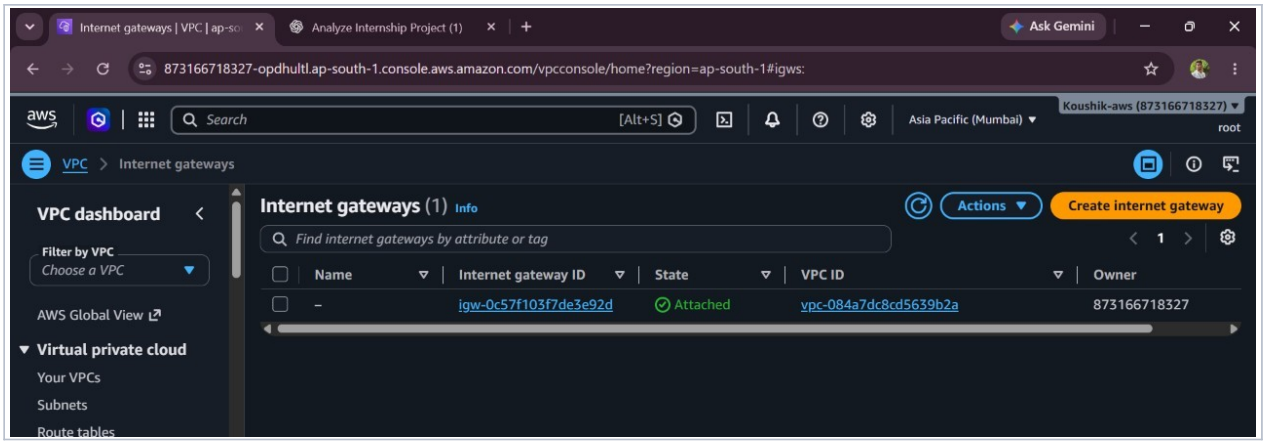


Figure 30. Cleanup, internet gateways: only igw-0c57f103f7de3e92d, attached to vpc-084a7dc8cd5639b2a, is listed. The project gateway igw-0fafab354c6908f0d is not listed.

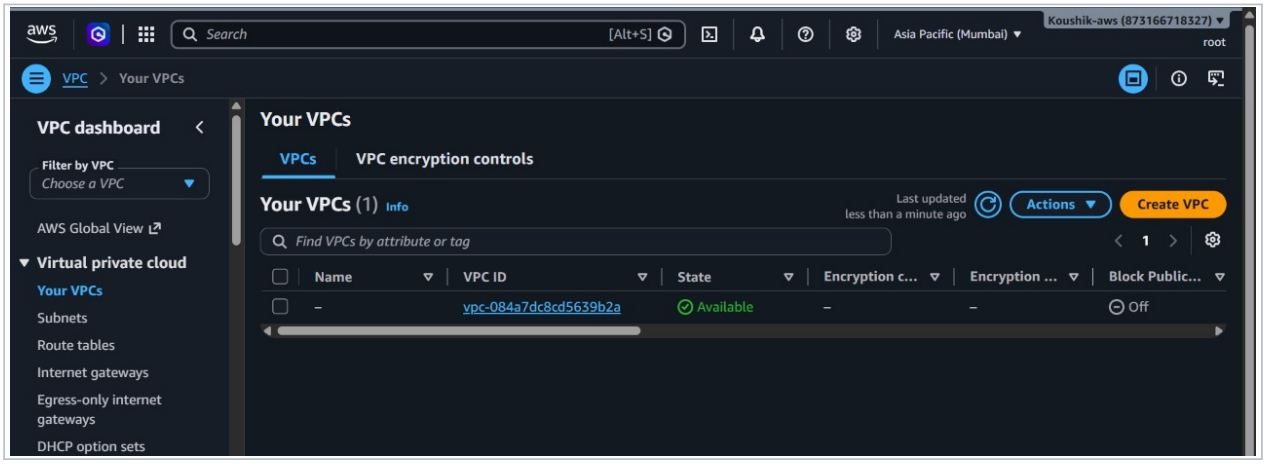


Figure 31. Cleanup, VPCs: only vpc-084a7dc8cd5639b2a (the default VPC) is listed. Neither project VPC is listed.

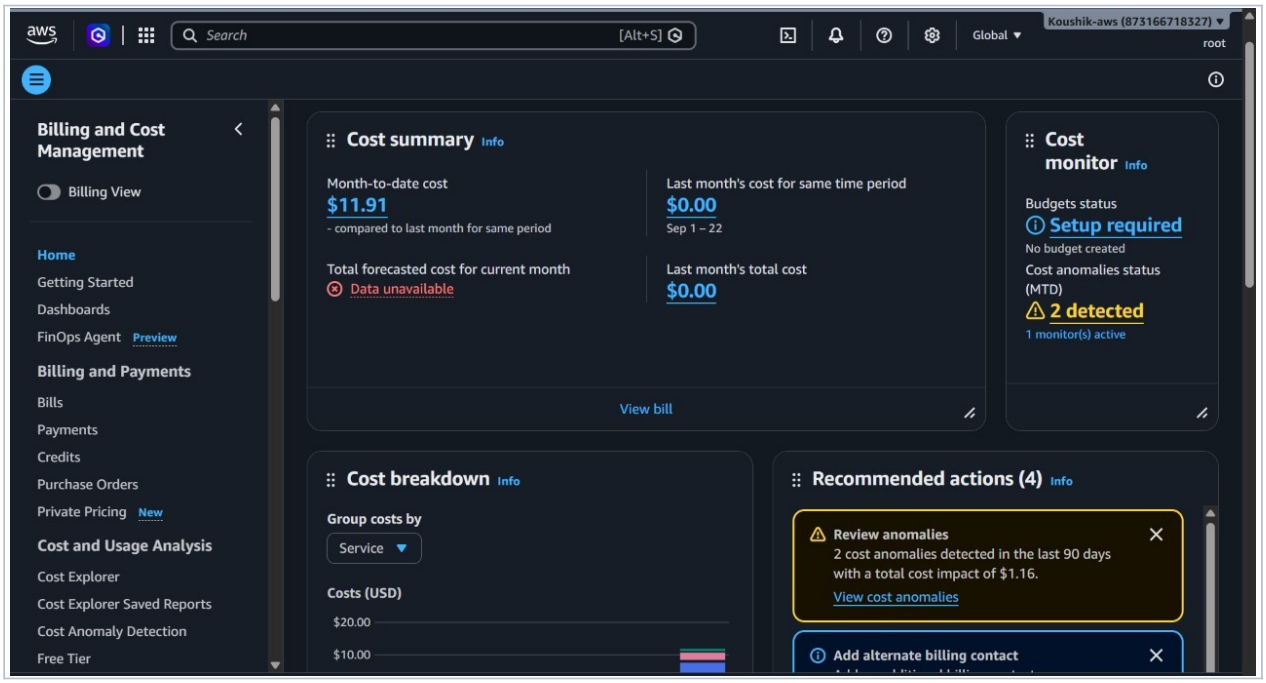


Figure 32. Billing home: month-to-date cost $11.91, last month $0.00, forecast 'Data unavailable', Budgets 'Setup required', 2 cost anomalies detected.

12 / REFLECTION

What I learned and conclusion

My notes on the project.

Learning note

During this internship, I gained practical experience in AWS networking and learned how different cloud resources work together. Working on the VPC Peering project helped me understand how to connect two separate Virtual Private Clouds and enable communication between instances using their IP addresses inside the VPCs.

I also learned the importance of route tables, security groups, and network rules in controlling communication between resources. Testing connectivity through ping and SSH helped me understand how to troubleshoot network issues and identify configuration mistakes.

Overall, this project gave me a better understanding of AWS networking beyond theoretical concepts. It also taught me the importance of checking configurations carefully, testing connectivity step by step, and cleaning up resources after completing a lab. This experience has increased my confidence in working with cloud technologies and motivated me to learn more about AWS and cloud infrastructure.

Observations from my setup

- Every rule uses a single /32 address, so the security groups allow only the traffic the lab needs.
- My security group screenshots show the edit page, not the saved rules, so I cannot show the saved state.
- The peering routes appear in all three route tables, and each route table has an explicit subnet association.
- I did not capture a reverse ping, an SSH login to ewb-app-b, or a controlled ICMP failure and recovery. Next time I will capture each step as I do it.
- The Billing page shows a month-to-date cost of $11.91 after the cleanup. It does not show what the cost is for, so I would check the Cost Explorer breakdown.

Conclusion

I connected two VPCs in ap-south-1 with an Active peering connection and added routes on both sides. My setup follows the workbook design: a bastion with a public IPv4 address in the VPC A public subnet, and two application instances with no public IP, one in each VPC. A ping from ewb-app-a to ewb-app-b worked (4 of 4 replies), and I logged in by SSH to the private instance ewb-app-a through the bastion. I did not capture a reverse ping, an SSH login to ewb-app-b, or a controlled ICMP failure and recovery. In the cleanup screenshots the instances are terminated, the peering connection is deleted, and only the default VPC resources are listed. The remaining gaps are listed in Appendix B.

APPENDIX A / SCREENSHOT INDEX

Screenshot index

Original files, figure numbers and the sections they belong to.

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Fig 1	Diagram drawn for this report (not a screenshot)	2
Fig 2	01-vpc-list.png	3
Fig 3	02-vpc-a.jpg	3
Fig 4	03-vpc-b.jpg	3
Fig 5	04-subnets-list.jpg	4
Fig 6	05-subnet-a-private_details.jpg	4
Fig 7	06-subnet-a-public.jpg	4
Fig 8	07-subnet-b-private.jpg	4
Fig 9	09-route-a-public.jpg	4
Fig 10	10-route-a-private.jpg	4
Fig 11	11-route-b-private.jpg	4
Fig 12	12-route-a-privte_subnet_ass.jpg	4
Fig 13	13-VPC-peering_con.jpg	5
Fig 14	14-inbound_security_group.jpg	6
Fig 15	15-inbound_security-a.jpg	6
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Fig 26	Cleanup screenshot 5: ewb-rt-a-public routes (pasted image)	11
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Fig 28	Cleanup screenshot 7: security groups (pasted image)	11
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Fig 31	Cleanup screenshot 10: VPCs (pasted image)	11
Fig 32	Cleanup screenshot 11: Billing home (pasted image)	11

The cleanup screenshots (Figures 22–32) were pasted into the chat without file names, so they are listed by what they show.

APPENDIX B / COVERAGE

Workbook checklist and remaining gaps

A clear record of what the evidence shows and what it does not show.

Workbook item	Status	Notes
Cover: name, batch, title, region and date	Complete	Included on the cover page.
Architecture diagram and resource inventory	Complete with limitation	The internet gateway name and the security group names are not visible in the screenshots.
Subnet CIDRs and route tables	Complete	Three subnets and three route tables with routes and associations are shown.
Active peering and security-group rules	Complete with limitation	Peering status is Active. The peering CIDRs and DNS settings are not visible. The security group screenshots show the edit page, with no group names or outbound rules.
Baseline failure before peering	Not captured	No ping or SSH test from before the peering routes were added.
Forward/reverse ping and SSH to the private host	Partly complete	Forward ping from ewb-app-a to ewb-app-b and SSH to ewb-app-a through the bastion are shown. The reverse ping and SSH to ewb-app-b are not captured.
Controlled ICMP failure and recovery	Not captured	No screenshots of a failed ping, a rule change or a recovery. Figure 20 shows the session where the ping was started, with no failed result.
Cleanup and learning note	Partly complete	The learning note is included. Cleanup evidence covers the instances, Elastic IPs, peering connection, subnets, security groups, route tables, internet gateway and VPCs. Volumes, snapshots and key pairs are not captured. Billing shows $11.91 month-to-date.

Final submission note

The lab resources have already been cleaned up, so the items marked ‘Not captured’ cannot be added without recreating them.